

Standard Self attention:

$$\text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Shape: Q (Seq, d_k)
 K (Seq, d_k)
 V (Seq, d_v)

Computational Complexity: $QK^T: O(S^2)$

$(QK^T)V: O(Sd_v)$

after this

Sequence length could be long

d_v is fixed (relatively small, e.g. 1024)

Key problem: QK^T

Solution: Linear Self-attention

$$\frac{\phi(Q)(\phi(K)V)}{\phi(Q)(\phi(K)\mathbf{1})}$$

1. We want to multiply K and V first:

$$(Q \cdot K) \cdot V \rightarrow Q(K \cdot V)$$

$$O(S^2) + O(Sd_v)$$

$$= O(S^2)$$

$$O(Sd_v) + O(Sd_v^2)$$

$$= O(S)$$

2. the ϕ is feature mapping. (note here we cannot use Softmax here because we times K and V first.)

However, we want to mimic softmax using ϕ

Common choice: $\phi(x) = \text{ELU}(x) + 1$

$$\text{ELU}(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha \exp(x) - 1 & \text{otherwise} \end{cases}$$

normally $\alpha = 1$

Why? $\rightarrow$ 1. force all values to be positive

$\rightarrow$ 2. introduce non-linearity (Softmax is non-linear)

$\rightarrow$ 3. Compute with Softmax: $e^{z_i \cdot K_j}$

$$\phi(Q)\phi(K) = e^{z_i} \cdot e^{K_j} = e^{z_{i,1}K_{j,1}} + \dots + e^{z_{i,d}K_{j,d}}$$

3. $\phi(Q)(\phi(K)\mathbf{1})$

$$= (\phi(Q)(\phi(K)\mathbf{1}))$$

$$= \sum_j \phi(Q_i) \phi(K_j)$$